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Apparatus and method for image analysis of a fluid sample on a test card to determine whether the fluid sample tests positive or negative for a bacterium or virus

Abstract

An apparatus and method for image analysis of a fluid sample on a test card are disclosed. An example apparatus for analyzing a fluid sample includes a camera imaging device configured to record a plurality of images of the fluid sample while located within a target zone of a fluid microchannel of a test card. The example apparatus also includes a controller configured to perform a cytometry analysis on the fluid sample to determine whether the fluid sample tests positive or negative for a bacteria or virus.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5147606	12/1991	Charlton et al.	N/A	N/A
5229297	12/1992	Schnipelsky et al.	N/A	N/A
5726026	12/1997	Wilding et al.	N/A	N/A
5976336	12/1998	Dubrow et al.	N/A	N/A
6068752	12/1999	Dubrow et al.	N/A	N/A
6071478	12/1999	Chow	N/A	N/A
6114122	12/1999	Besemer et al.	N/A	N/A
6153073	12/1999	Dubrow et al.	N/A	N/A
6235175	12/2000	Dubrow et al.	N/A	N/A
6303343	12/2000	Kopf-Sill	N/A	N/A
6399025	12/2001	Chow	N/A	N/A
6428987	12/2001	Franzen	N/A	N/A
6437551	12/2001	Krulevitch et al.	N/A	N/A
6509186	12/2002	Zou et al.	N/A	N/A
6576459	12/2002	Miles et al.	N/A	N/A
7033474	12/2005	Dubrow et al.	N/A	N/A
7309467	12/2006	Chen et al.	N/A	N/A
7431888	12/2007	Frechet et al.	N/A	N/A
7678336	12/2009	Chang et al.	N/A	N/A
7811523	12/2009	Bjorneson	N/A	N/A
7867754	12/2010	Regnier et al.	N/A	N/A
7883669	12/2010	Sun et al.	N/A	N/A

7915030	12/2010	Inoue et al.	N/A	N/A
7919062	12/2010	Yuen	N/A	N/A
7981237	12/2010	Park et al.	N/A	N/A
8053239	12/2010	Wheeler et al.	N/A	N/A
8158926	12/2011	Feng et al.	N/A	N/A
8202491	12/2011	Masters et al.	N/A	N/A
8216827	12/2011	Pouteau et al.	N/A	N/A
8247176	12/2011	Pourahmadi et al.	N/A	N/A
8289519	12/2011	Zare et al.	N/A	N/A
8343778	12/2012	Li et al.	N/A	N/A
8349276	12/2012	Paluma et al.	N/A	N/A
8367021	12/2012	Kennedy et al.	N/A	N/A
8394341	12/2012	Reinhardt et al.	N/A	N/A
8409848	12/2012	Zeng et al.	N/A	N/A
8540946	12/2012	Padmanabhan et al.	N/A	N/A
8557199	12/2012	Heath et al.	N/A	N/A
8557518	12/2012	Jovanovich et al.	N/A	N/A
8562918	12/2012	Jovanovich et al.	N/A	N/A
8592157	12/2012	Pourahmadi et al.	N/A	N/A
8597574	12/2012	Gumbrecht et al.	N/A	N/A
8603414	12/2012	Omuro et al.	N/A	N/A
8606414	12/2012	Ludwig	N/A	N/A
8790595	12/2013	Polwart et al.	N/A	N/A
8852527	12/2013	Thomas et al.	N/A	N/A
8874273	12/2013	Sun et al.	N/A	N/A
8894946	12/2013	Nielsen et al.	N/A	N/A
8911636	12/2013	Gautham	N/A	N/A
8911989	12/2013	Lee et al.	N/A	N/A
8936762	12/2014	Ehrlich et al.	N/A	N/A
8940147	12/2014	Bartsch et al.	N/A	N/A
8962252	12/2014	Liang et al.	N/A	N/A
9017946	12/2014	Hasson	N/A	N/A
9114398	12/2014	Knight et al.	N/A	N/A
9138744	12/2014	Tsao et al.	N/A	N/A
9170138	12/2014	Giovangrandi et al.	N/A	N/A
9328344	12/2015	Link et al.	N/A	N/A
9335247	12/2015	Sharpe et al.	N/A	N/A
9364833	12/2015	Bergstedt	N/A	N/A
9540686	12/2016	Zeng et al.	N/A	N/A
10761019	12/2019	Khodadad et al.	N/A	N/A
11371091	12/2021	Revilla et al.	N/A	N/A
11692967	12/2022	Koussa et al.	N/A	N/A
2001/0029793	12/2000	Moler et al.	N/A	N/A
2001/0045355	12/2000	Gephart et al.	N/A	N/A
2003/0082568	12/2002	Phan et al.	N/A	N/A
2003/0155344	12/2002	Cobb	N/A	N/A
2003/0190608	12/2002	Blackburn	N/A	N/A
2004/0037739	12/2003	McNeely et al.	N/A	N/A
2004/0193202	12/2003	Allen	N/A	N/A
2005/0032204	12/2004	Rodgers et al.	N/A	N/A

2005/0196779	12/2004	Ho et al.	N/A	N/A
2005/0233440	12/2004	Scurati et al.	N/A	N/A
2006/0013725	12/2005	Larsen	N/A	N/A
2006/0094028	12/2005	Danna et al.	N/A	N/A
2007/0015179	12/2006	Klapperich	N/A	N/A
2007/0154895	12/2006	Spaid et al.	N/A	N/A
2007/0163175	12/2006	Kihara et al.	N/A	N/A
2007/0190828	12/2006	Goldman et al.	N/A	N/A
2007/0259348	12/2006	Phadke et al.	N/A	N/A
2008/0200343	12/2007	Clemens et al.	N/A	N/A
2008/0241910	12/2007	Jung et al.	N/A	N/A
2008/0253633	12/2007	Xia et al.	N/A	N/A
2009/0004732	12/2008	LaBarre et al.	N/A	N/A
2009/0018640	12/2008	State	N/A	N/A
2009/0053726	12/2008	Owen et al.	N/A	N/A
2009/0062134	12/2008	Linton et al.	N/A	N/A
2009/0140170	12/2008	Nevill et al.	N/A	N/A
2009/0143233	12/2008	Knight et al.	N/A	N/A
2009/0186404	12/2008	Kim et al.	N/A	N/A
2009/0215157	12/2008	Jung et al.	N/A	N/A
2009/0215194	12/2008	Magni et al.	N/A	N/A
2009/0311717	12/2008	De Sonnevile et al.	N/A	N/A
2010/0203521	12/2009	Klapperich et al.	N/A	N/A
2010/0261286	12/2009	Kim et al.	N/A	N/A
2010/0291584	12/2009	Tseng et al.	N/A	N/A
2010/0304986	12/2009	Chen et al.	N/A	N/A
2011/0020876	12/2010	Wilding et al.	N/A	N/A
2011/0039280	12/2010	Leary et al.	N/A	N/A
2011/0117673	12/2010	Johnson et al.	N/A	N/A
2011/0162439	12/2010	Ayliffe	N/A	N/A
2011/0206545	12/2010	Junod et al.	N/A	N/A
2011/0269131	12/2010	Chiu et al.	N/A	N/A
2011/0301535	12/2010	Takayama et al.	N/A	N/A
2011/0315559	12/2010	Holt et al.	N/A	N/A
2012/0052560	12/2011	Knight et al.	N/A	N/A
2012/0140055	12/2011	Narusawa et al.	N/A	N/A
2012/0145253	12/2011	Zeng et al.	N/A	N/A
2012/0195810	12/2011	Cohen et al.	N/A	N/A
2012/0220047	12/2011	Seifried et al.	N/A	N/A
2012/0244043	12/2011	LeBlanc et al.	N/A	N/A
2012/0244604	12/2011	Kornilovich	N/A	N/A
2012/0283108	12/2011	Sampas	N/A	N/A
2012/0309010	12/2011	Shuber	N/A	N/A
2013/0052725	12/2012	Yazdanfar	N/A	N/A
2013/0149215	12/2012	Dekker et al.	N/A	N/A
2013/0224781	12/2012	Jung et al.	N/A	N/A
2013/0236907	12/2012	Petersen et al.	N/A	N/A
2013/0244906	12/2012	Collins	N/A	N/A
2013/0345096	12/2012	Wan	N/A	N/A
2014/0038191	12/2013	Liang et al.	N/A	N/A

2014/0141424	12/2013	Pourahmadi et al.	N/A	N/A
2014/0161686	12/2013	Bort et al.	N/A	N/A
2014/0162893	12/2013	Cash et al.	N/A	N/A
2014/0170672	12/2013	Vandersleen et al.	N/A	N/A
2014/0199764	12/2013	Domansky et al.	N/A	N/A
2014/0200154	12/2013	Sugarman et al.	N/A	N/A
2014/0248621	12/2013	Collins	N/A	N/A
2014/0295441	12/2013	Egan et al.	N/A	N/A
2014/0307931	12/2013	Gierahn et al.	N/A	N/A
2014/0309508	12/2013	Kim et al.	N/A	N/A
2015/0024426	12/2014	De Oliveira Garcia Da Fonseca et al.	N/A	N/A
2015/0125947	12/2014	Korczyk et al.	N/A	N/A
2015/0238967	12/2014	Erickson et al.	N/A	N/A
2015/0290644	12/2014	Prentice et al.	N/A	N/A
2016/0033311	12/2015	Giovangrandi et al.	N/A	N/A
2016/0069913	12/2015	Bakhru et al.	N/A	N/A
2016/0144356	12/2015	Jung et al.	N/A	N/A
2016/0144358	12/2015	Patel	N/A	N/A
2016/0193603	12/2015	Battrell et al.	N/A	N/A
2016/0296933	12/2015	Chiou et al.	N/A	N/A
2016/0310948	12/2015	Nowakowski et al.	N/A	N/A
2016/0318019	12/2015	Ledden et al.	N/A	N/A
2016/0334351	12/2015	Lu et al.	N/A	N/A
2016/0340716	12/2015	Ortac et al.	N/A	N/A
2016/0369323	12/2015	Revilla et al.	N/A	N/A
2017/0001196	12/2016	Zhang et al.	N/A	N/A
2017/0008009	12/2016	Azpiroz et al.	N/A	N/A
2017/0021354	12/2016	Kim et al.	N/A	N/A
2017/0058324	12/2016	Balog et al.	N/A	N/A
2018/0015475	12/2017	Arlett et al.	N/A	N/A
2018/0174689	12/2017	Pulitzer et al.	N/A	N/A
2019/0160460	12/2018	Keatch et al.	N/A	N/A
2020/0064254	12/2019	Vanderklein et al.	N/A	N/A
2020/0383664	12/2019	Loudermilk et al.	N/A	N/A
2021/0299651	12/2020	McCord et al.	N/A	N/A
2021/0311042	12/2020	Pamula et al.	N/A	N/A
2021/0389313	12/2020	Ward et al.	N/A	N/A
2022/0088583	12/2021	Kleinemolen et al.	N/A	N/A
2022/0411851	12/2021	Revilla et al.	N/A	N/A
2023/0116264	12/2022	Basmajian et al.	N/A	N/A
2023/0175964	12/2022	Taylor et al.	N/A	N/A
2023/0194540	12/2022	Basmajian et al.	N/A	N/A
2023/0228674	12/2022	Marrinucci et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202018102600	12/2017	DE	N/A
2863332	12/2014	EP	N/A
93/22053	12/1992	WO	N/A

1993022054	12/1992	WO	N/A
01/54813	12/2000	WO	N/A
01/73420	12/2000	WO	N/A
2013177953	12/2012	WO	N/A
2014055963	12/2013	WO	N/A
2014144548	12/2013	WO	N/A
2016209775	12/2015	WO	N/A
2017087834	12/2016	WO	N/A
2021226581	12/2020	WO	N/A
2022029731	12/2021	WO	N/A
2022051703	12/2021	WO	N/A
2022256514	12/2021	WO	N/A
2023059894	12/2022	WO	N/A

OTHER PUBLICATIONS

Boehm, Douglas A., Philip A. Gottlieb, and Susan Z. Hua. "On-chip microfluidic biosensor for bacterial detection and identification." *Sensors and Actuators B: Chemical* 126.2 (2007): 508-514. cited by examiner

Wolfe, Alan J., et al. "Evidence of uncultivated bacteria in the adult female bladder." *Journal of clinical microbiology* 50.4 (2012): 1376-1383. cited by examiner

International Search Report and Written Opinion issued in related International Patent Application No. PCT/US2016/038152 on Sep. 14, 2016. 12 pages. cited by applicant

International Search Report and Written Opinion issued in related International Patent Application No. PCT/US2016/038124 on Sep. 8, 2016. 12 pages. cited by applicant

International Search Report and Written Opinion issued in related International Patent Application No. PCT/US2016/038157 on Oct. 28, 2016. 12 pages. cited by applicant

Liu, Hao-Bing et al., Micro air bubble formation and its control during polymerase chain reaction (PCR) in polydimethylsiloxane (PDMS) microreactors:, *Journal of Micromechanics and Microengineering* 17.10 (2007): 2055. cited by applicant

Cady, Nathaniel C., et al. "Real-time PCR detection of *Listeria monocytogenes* using an integrated microfluidics platform." *Sensors and Actuators B: Chemical* 107.1 (2005): 332-341. cited by applicant

Jarvis, Jonas, et al. "Digital quantification using amplified single-molecule detection." *Nature methods* 3.9 (2006): 725. cited by applicant

Lui, Clarissa, Nathaniel Cady, and Carl Batt. "Nucleic acid-based detection of bacterial pathogens using integrated microfluidic platform systems." *Sensors* 9.5 (2009): 3713-3744. cited by applicant

International Preliminary Report on Patentability issued in related International Patent Application No. PCT/US2016/038152 on Jun. 28, 2017. cited by applicant

Miralles et al., "A Review of Heating Temperature Control in Microfluidic Systems: Techniques and Applications," *Diagnostics*, 2013, No. 3, pp. 33-67. cited by applicant

Wang et al., A miniaturized quantitative polymerase chain reaction system for DNA amplification and detection: *Sensors and Actuators*, No. B 141, 2009, pp. 329-337. cited by applicant

Hsieh et al., "Enhancement of thermal uniformity for a microthermal cyler and its application for polymerase chain reaction," *Sensors and Actuators*, B 130 (2008), pp. 848-856. cited by applicant

International Preliminary Report on Patentability issued in related international Patent Application No. PCT/US2016/038124 on Jun. 26, 2017. cited by applicant

International Preliminary Report on Patentability issued in related international Patent Application No. PCT/US2016/038157 on Jun. 19, 2017. cited by applicant

Creative Materials Product Description—Dielectric Inks and Coatings Offerings (2015), 2 pages. cited by applicant

Extended Search Report for EP 16815096.9, dated Mar. 7, 2019, 4 pages. cited by applicant
Partial Search Report for EP 23161706.9, dated Aug. 4, 2023, 13 pages. cited by applicant
Partial Search Report for EP 23162636.7, dated Apr. 16, 2024, 12 pages. cited by applicant
International Search Report and Written Opinion for PCT/US2021/065815, mailed Mar. 24, 2022, 12 pages. cited by applicant
Gubala et al., "Point of Care Diagnostics: Status and Future," Analytical Chemistry, 2012, vol. 84, No. 2, pp. 487-515. cited by applicant

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE TO RELATED APPLICATIONS (1) This non-provisional patent application is a continuation of U.S. patent application Ser. No. 15/185,714, entitled "Apparatus and Method for Image Analysis of a Fluid Sample Undergoing a Polymerase Chain Reaction ("PCR")", filed Jun. 17, 2016, which claims priority to and the benefit of U.S. Provisional Patent Application No. 62/182,992, entitled "Point-Of-Care PCR Assay for Infectious Agents", filed Jun. 22, 2015, and U.S. Provisional Patent Application No. 62/187,471, entitled "Point-Of-Care PCR Assay for Infectious Agents", filed Jul. 1, 2015, the entire contents of each of which are hereby incorporated by reference and relied upon. (2) This application is related to U.S. application Ser. No. 15/185,640, entitled "Device for Analyzing a Fluid Sample and Use of Test Card with Same", filed Jun. 17, 2016, and U.S. application Ser. No. 15/185,661, now U.S. Pat. No. 10,214,772, entitled "Test Card for Assay and Method of Manufacturing Same", filed Jun. 17, 2016, the entire contents of each of which are hereby incorporated by reference and relied upon.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to apparatuses and methods for performing an assay, and more specifically to apparatuses and methods for performing image analysis of a fluid sample on a test card.

BACKGROUND

(2) Point-of-care (POC) in vitro diagnostics tests (IVDT) have traditionally had two major categories, nucleic acid amplification tests (NAAT) or immunoassay-based tests. The former directly detects the pathogen's DNA or RNA, while the latter detects antibodies or antigens generated by the immune system response to the pathogen.

(3) Current POC diagnostic immunoassays lack the high sensitivity and specificity of nucleic acid amplification methods. This becomes more pronounced during the initial stages of infection, often within 168 hours. Taking the case of Dengue virus in whole blood, immunoglobulin M (IgM) and immunoglobulin G (IgG) remain undetectable in the majority of patients until 5 and 10 days post-infection, respectively, whereas nucleic acid can be found as early as 0 to 7 days. Moreover, many immunoassay tests are unable to detect infectious agents until 3 months after the initial onset of the infection. This delay is due to the time it takes for the body's immune system to respond to an infection.

(4) POC diagnostic assays developed utilizing NAATs have very high sensitivities and specificities, matching those of currently accepted laboratory tests. The primary mechanism of NAAT based systems is to directly detect an infectious agent's nucleic acid, lending to the test's ability to detect diseases within the first few days of the onset of infection. In addition, by careful primer design, NAATs also have the ability to have very high specificity and sensitivity compared to immunoassay

based testing. The largest drawback of NAATs compared to immunoassay-based tests is the complicated equipment and/or processes required to prepare a sample for testing.

SUMMARY

(5) Described herein are methods and apparatus for causing a point-of-care polymerase chain reaction and analyzing the polymerase chain reaction at the point-of-care, particularly when unwanted bubbles are present during the polymerase chain reaction. In a general embodiment, a device for analyzing a polymerase chain reaction in a fluid sample includes a current source configured to cause the polymerase chain reaction by heating the fluid sample within a target zone, a camera imaging device configured to record a plurality of images of the fluid sample in the target zone while the current source causes the polymerase chain reaction, and a controller configured to (i) distinguish wanted objects in the plurality of images from an unwanted object in the plurality of images, and (ii) determine whether the fluid sample tests positive or negative for a bacteria or virus based on the wanted objects.

(6) In an example embodiment, the wanted objects are nucleic acid molecules and the unwanted object is an air bubble.

(7) In an example embodiment, the controller is configured to analyze the plurality of images by dividing the target zone into a plurality of bins.

(8) In an example embodiment, the plurality of bins are arranged in a grid with a plurality of rows and columns.

(9) In an example embodiment, the controller is configured to determine whether the fluid sample has tested positive or negative for the bacteria or virus by selecting at least two of the plurality of bins that overlap a cluster of wanted objects and calculating a mean fluorescence value of the at least two of the plurality of bins.

(10) In an example embodiment, the controller is configured to exclude at least one of the plurality of bins from the mean fluorescence value calculation if the at least one of the plurality of bins overlaps the unwanted object.

(11) In an example embodiment, the controller is configured to assign a weight to at least one of the plurality of bins used in the mean fluorescence value calculation based on the proximity of the at least one of the plurality of bins to the unwanted object.

(12) In an example embodiment, the controller is configured to exclude at least one of the plurality of bins if the at least one of the plurality of bins does not meet a minimum threshold value for brightness.

(13) In an example embodiment, the controller is configured to report an inconclusive test if the controller identifies an unwanted image in the plurality of images.

(14) In an example embodiment, the device includes a user interface, the controller includes a plurality of preprogrammed analyses that can be performed on the fluid sample, and the user interface is configured to allow a user to select at least one analysis from the plurality of preprogrammed analyses.

(15) In an example embodiment, the plurality of images includes at least one of: (i) a plurality of still images of the polymerase chain reaction recorded by the camera imaging device over a period of time; or (ii) a video image of the polymerase chain reaction recorded by the camera imaging device over the period of time.

(16) In an example embodiment, the controller is configured to distinguish the wanted objects in the plurality of images from the unwanted object in the plurality of images by comparing a size or shape of objects in the plurality of images to an average size or shape of blood cells.

(17) In a general embodiment, a device for analyzing a polymerase chain reaction in a fluid sample includes a current source configured to cause the polymerase chain reaction by heating the fluid sample within a target zone, a camera imaging device configured to record a plurality of images of the fluid sample in the target zone while the current source causes the polymerase chain reaction, and a controller configured to analyze the plurality of images by (i) dividing the target zone shown

in the plurality of images into a plurality of bins, (ii) selecting at least two of the plurality of bins, and (iii) calculating a fluorescence value of the selected at least two of the plurality of bins.

(18) In an example embodiment, the controller is configured to select the at least two of the plurality of bins based on a threshold value for brightness.

(19) In an example embodiment, the controller is configured to exclude at least one of the plurality of bins from the fluorescence value calculation if the at least one of the plurality of bins overlaps an unwanted object.

(20) In an example embodiment, the controller is configured to assign weights to the selected at least two of the plurality of bins for the calculation based on the proximity of the selected at least two of the plurality of bins to an unwanted object.

(21) In a general embodiment, a method of analyzing a polymerase chain reaction in a fluid sample includes heating the fluid sample in a target zone to cause the polymerase chain reaction, recording a plurality of images of the fluid sample in the target zone during the polymerase chain reaction, dividing the target zone into a plurality of bins, calculating a fluorescence value of at least two of the plurality of bins, and determining whether the fluid sample tests positive or negative for a bacteria or virus based on the calculated fluorescence value.

(22) In an example embodiment, calculating the fluorescence value includes weighting the at least two of the plurality of bins based on proximity to an unwanted object.

(23) In an example embodiment, calculating the fluorescence value includes excluding at least one of the plurality of bins from the fluorescence value calculation if the at least one of the plurality of bins overlaps an unwanted object.

(24) In an example embodiment, calculating the fluorescence value includes excluding at least one of the plurality of bins based on a threshold value for brightness.

(25) In a general example embodiment, a device for analyzing a fluid sample includes a camera imaging device configured to record a plurality of images of the fluid sample while located within a target zone of a fluid microchannel, and a controller configured to (i) distinguish wanted objects in the plurality of images from an unwanted object in the plurality of images, and (ii) determine whether the fluid sample tests positive or negative for a bacteria or virus based on the wanted objects.

(26) In an example embodiment, the controller is configured to determine whether the fluid sample tests positive or negative for the bacteria or virus by dividing the target zone shown in the plurality of images into a plurality of bins, selecting at least two of the plurality of bins, and calculating a fluorescence value of the selected at least two of the plurality of bins.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the present disclosure will now be explained in further detail by way of example only with reference to the accompanying figures, in which:
- (2) FIG. 1 is a top perspective view of an example embodiment of an assay device according to the present disclosure;
- (3) FIG. 2 is a top perspective view of an example embodiment of a test card according to the present disclosure;
- (4) FIG. 3 is a cross-sectional view of the test card of FIG. 2;
- (5) FIG. 4 is a top perspective view of the assay device of FIG. 1 with the test card of FIG. 2 inserted therein;
- (6) FIG. 5 is a cross-sectional view of the assay device and test card shown in FIG. 4;
- (7) FIGS. 6A to 6C illustrate an example embodiment of amplified images showing a PCR;
- (8) FIGS. 7A to 7E illustrate an example embodiment showing how a controller can detect bubbles

and analyze a reaction even when bubbles are detected;

(9) FIG. 8 illustrates an example embodiment of a control method that can be used to perform and analyze a reaction according to the present disclosure;

(10) FIG. 9 illustrates an example embodiment of a controller that can perform the method of FIG. 8.

DETAILED DESCRIPTION

(11) Before describing in detail the illustrative system and method of the present disclosure, it should be understood and appreciated herein that the present disclosure relates to a rapid, high sensitivity and high specificity, low complexity, diagnostic system **1** using nucleic acid amplification and capable of operating in low resource settings with minimal user training. The system described herein is configured, for example, to cause and analyze polymerase chain reactions (PCR), particularly in the early stages of infection, using a low-cost microfluidic platform employing PCR with a modified DNA polymerase.

(12) FIG. 1 illustrates an example embodiment of a point-of-care diagnostic system **1** according to the present disclosure. As illustrated, diagnostic system **1** includes an assay device **10** with a housing **12** having a slot **14** to receive a test card **100** (FIG. 2), which is an inexpensive, disposable test card that can be used with device **10** and then discarded. As explained in more detail below, a fluid sample can be injected into test card **100**, and then test card **100** can be inserted into slot **14** so that device **10** can power test card **100** to run an assay within test card **100** without further action by the user. The resulting analysis can then be displayed to the user by user interface **60**. Test card **100** can then be discarded and a new test card **100** can be inserted into slot **14** and used the same way to run a new assay. In an embodiment, the test card is configured to receive about 10 μ L of whole blood, the equivalent to a drop of blood obtained from a finger stick. In another embodiment, the fluid sample can be serum, urine, saliva, tears and/or the like.

(13) Device **10** is described in more detail in U.S. application Ser. No. 15/185,640, entitled “Device for Analyzing a Fluid Sample and Use of Test Card with Same”, filed Jun. 17, 2016, the entire disclosure of which, and specifically the device and test card structure disclosure, is incorporated herein by reference and relied upon. Those of ordinary skill in the art will recognize other configurations of device **10** that can be used according to the present disclosure.

(14) Test card **100** is described in more detail in U.S. application Ser. No. 15/185,661, now U.S. Pat. No. 10,214,772, entitled “Test Card for Assay and Method of Manufacturing Same”, filed Jun. 17, 2016, the entire disclosure of which, and specifically the test card structure disclosure, is incorporated herein by reference and relied upon. Those of ordinary skill in the art will recognize other configurations of test card **100** that can be used according to the present disclosure.

(15) As illustrated in FIGS. 2 and 3, test card **100** includes an inlet port **124**, a mixing chamber **126**, a capture port **128**, an outlet port **130**, and a fluid microchannel **134**. A liquid sample can be injected into inlet port **124** and mixed with one or more reagent in mixing chamber **126**, and then test card **100** can be placed into slot **14** of assay device **10**. Once test card **100** has been placed within device **10**, the fluid sample can be pulled through fluid microchannel **134**, so that the fluid sample can be analyzed through an analysis port **132** of test card **100**. Test card **100** also includes electrical contacts **122** on a bottom surface **102** thereof, which enables electrodes adjacent to fluid microchannel **134** to be controlled to power test card **100**, for example, to heat fluid within fluid microchannel **134**, track fluid flow through fluid microchannel **134** and/or measure the concentration of a chemical species in the fluid sample.

(16) FIG. 4 illustrates a perspective view of device **10** after test card **100** has been placed into slot **14**, and FIG. 5 illustrates a cross-sectional view thereof. As illustrated, placement of test card **100** into slot **14** aligns several of the elements of device **10** with several of the elements of test card **100**. For example, placement of test card **100** into slot **14** aligns camera imaging device **20** and light source **22** of device **10** with analysis port **132** on an upper surface **104** of test card **100**, pneumatic tube **40** of fluid actuation source **24** with outlet port **130** on the upper surface **104** of test

card **100**, and electrical contacts **42** of electrical contact device **26** with electrical contacts **122** on the bottom surface **102** of test card **100**.

(17) As explained in more detail in U.S. application Ser. No. 15/185,640, once pneumatic tube **40** is sealed against outlet port **130**, a negative pneumatic force can be applied to outlet port **130** from fluid actuation source **24**. When the negative pneumatic force is applied, the fluid sample injected into inlet port **124** is pulled through fluid microchannel **134** towards outlet port **130**. The fluid sample however is not pulled into pneumatic tube **40** due to the presence of capture port **128** between inlet port **124** and outlet port **130**. Capture port **128** allows fluid to build up before it can reach outlet port **130** and/or pneumatic tube **40**, which keeps device **10** sterile and protects the integrity of diagnostic system **1**.

(18) Test card **100** is dimensioned so that electrical contacts **122** of test card **100** are placed into electrical contact with electrical contacts **42** of electrical contact device **26** when test card **100** is fully inserted into slot **14**. With the electrical contacts **42** and **122** aligned, controller **28** can perform several functions. Before beginning an assay, controller **28** can ensure that the fluid sample has been properly pulled through microchannel **134** and into a target zone of microchannel **134** by measuring the capacitance of the fluid sample upstream and/or downstream of the target zone. Once it is determined that fluid is located within the target zone of microchannel **134**, controller **28** can control power source **30** to apply a current to electrical contacts aligned with the target zone to heat the fluid sample located within target zone and cause a reaction such as a PCR to occur.

(19) The alignment of camera imaging device **20** over analysis port **132** on the upper surface **104** of test card **100** allows controller **28** to analyze one or more reaction within fluid microchannel **134** while controlling the reaction. Camera imaging device **20** is configured to record a plurality of images of the fluid sample within the target zone so that the images can be analyzed in real time. The plurality of images can include, for example, a plurality of still images and/or a video image. Camera imaging device **20** is therefore configured to record the reaction of the fluid sample by taking a series of still images of the fluid sample within the target zone during a reaction or by taking a video of the fluid sample within the target zone during a reaction. As explained in more detail below, the plurality of images can be used, for example, to analyze the physical and/or chemical characteristics of cells within the fluid sample.

(20) The target zone can be anywhere along fluid microchannel **134** or can be branched off of fluid microchannel **134**. Example embodiments of target zones are described in more detail in U.S. application Ser. No. 15/185,661, now U.S. Pat. No. 10,214,772, entitled “Test Card for Assay and Method of Manufacturing Same”, filed Jun. 17, 2016. In an embodiment, the target zone can be located at a central portion of fluid microchannel **134** and fluid microchannel **134** can include, for example, a capacitance sensor upstream and/or downstream of the target zone to determine whether fluid has been pulled through fluid microchannel **134** and into the target zone. In an embodiment, microchannel **134** can include a plurality of target zones to perform a plurality of assays at the same or different times, for example, to improve the reliability of the assays. In an embodiment, controller **28** can be programmed to analyze a specific section of microchannel **134** as the target zone, or controller **28** can determine the target zone based on the plurality of images taken by camera imaging device **20**.

(21) FIGS. **6A** to **6C** illustrate an example embodiment of amplified images showing a PCR. FIG. **6A** illustrates a fluid sample that has been initially loaded into target zone **166**, with a few nucleic acid molecules **50** separated by large distances. FIG. **6B** shows the fluid sample during the PCR, when clear colonies **52** of nucleic acid molecules **50** can be viewed because the diffusion speed of the molecules is slower than the speed required for the molecules to travel the distance between colonies. FIG. **6C** shows the end of the PCR, when the colonies **52** are no longer clearly visible.

(22) From the number of colonies shown in FIG. **6B**, controller **28** can determine whether the fluid sample tests positive or negative for a specific virus or bacteria based on a known titer value for the specific virus or bacteria. The titer value corresponds to the highest dilution factor for the specific

virus or bacteria that yields a positive test result. By counting the colonies in FIG. 6B, and by comparing the count with the known titer value, controller **28** can determine a positive or negative test result for the fluid sample. The known titer value can be compared through a variety of methods. In an embodiment, a statistical analysis can be utilized to say with certain probability that the result will fall under a certain titer value. This can be achieved by running a statistically significant amount of controlled samples under controlled dilutions and generate a standard curve based on the crossover threshold value of each (point at which the curve starts to increase).

(23) In an embodiment, camera imaging device **20** includes a high sensitivity and dynamic range complementary metal-oxide semiconductor (CMOS) camera sensor which allows for general imaging of a reaction within target zone **166** of fluid microchannel **134** of test card **100**. In an embodiment, the CMOS camera sensor enables camera imaging device **20** to image the entire analysis port **132** of test card **100**. Although in the illustrated embodiment only a single target zone **166** within a single microchannel **134** is shown, camera imaging device **20** is configured to image a plurality of target zones and/or a plurality of microchannels should the user insert such a test card **100** into device **10**. Examples of alternative embodiments of test cards **100** are disclosed in U.S. application Ser. No. 15/185,661, now U.S. Pat. No. 10,214,772, entitled “Test Card for Assay and Method of Manufacturing Same”, filed Jun. 17, 2016.

(24) In the illustrated embodiment, light source **22** is configured to project a fluorescent excitation light on target zone **166** of fluid microchannel **134** during a PCR, while camera imaging device **20** is recording still and/or video images of the PCR. In an embodiment, the fluid sample has been mixed with a fluorescence reagent in mixing port **126**, before the fluid sample is pulled into target zone **166**. By illuminating target zone **166** with a fluorescence excitation light, controller **28** can record fluorescence measurements from the images taken by camera imaging device **20**. The fluorescence images can be analyzed to determine the incremental increases in fluorescence during each PCR cycle. This allows the determination of the PCR amplification curve. Additionally, the images can be used during melt analysis to determine the incremental decrease in fluorescence as temperature is increased. Incremental change in fluorescence can be determined by taking the difference in pixel intensity between successive images.

(25) For general fluorescence measurements, for example, a statistical formulation of the PCR can be achieved by subdividing the target zone **166** and measuring fluorescence at various locations in the target zone **166**. This yields highly-accurate and consistent crossover threshold values for the real-time PCR based on titer values. Once the PCR is complete, a melting curve analysis allows for error checking, ensuring that the correct amplicon has been amplified during the PCR, to reduce the likelihood of a false-positive test.

(26) Camera imaging device **20** also enables controller **28** to record a variety of other measurements in addition to fluorescence measurements. For example, controller **28** can determine turbidity and object detection using the images taken by camera imaging device **20**. The turbidity can be determined by using the device LEDs to apply incident light to a test card microchannel and/or target zone. The amount of scattered light can be measured by the camera allowing a numerical value of relative turbidity to be measured. The turbidity can be used to determine the outcome of various different reactions, such as measuring the reactivity of proteins during an ELISA test (colorimetry measurement) or measurement of water quality. The object detection can be used, for example, to determine whether air bubbles formed in target zone **166** during the PCR and potentially destroyed the integrity of the PCR analysis. By imaging the entire analysis port **132** of test card **100**, camera imaging device **20** is able to greatly improve accuracy by allowing for various advanced image-processing algorithms to be applied.

(27) With respect to object detection, camera imaging device **20** can be used to detect both wanted and unwanted objects within microchannel **134**. In an embodiment, the wanted objects are cells such as red and white blood cells in the fluid sample to be analyzed during the reaction. Controller **28** can count the wanted objects by matching the cell size and cell shape of objects in the images

from camera imaging device **20** to the average size and shape of specific blood cells, allowing controller **28** to count, for example, the number of red and white blood cells in the fluid sample at target zone **166**. Controller **28** can also distinguish the red and white blood cells from other unwanted objects that do not match the average size and shape of specific blood cells.

(28) If controller **28** detects an unwanted object, controller **28** can determine that the PCR has failed or is indeterminate, and can instruct a user that an additional PCR should be run on the same or a different test card. If the PCR is run on the same test card, target zone **166** should be cleared of fluid, and then new fluid should be pulled from mixing chamber **126** into target zone **166**.

(29) In an embodiment, the unwanted image is an air bubble. In an embodiment, controller **28** can detect air bubbles because air bubbles will expand in the target zone **166** while the PCR reaction occurs. Red and white blood cells, on the other hand, multiply but do not increase in size. Controller **28** can therefore determine the presence of an air bubble in target zone **166** by noting a change in size of an object across a plurality of subsequent images from the plurality of images. In another embodiment, controller **28** can determine the presence of an air bubble by noting that an unwanted object does not match the average size and shape of blood cells. In another embodiment, controller **28** can determine the presence of an air bubble due to there being no fluorescence inside of the bubble by looking for a change in the gradient of fluorescence.

(30) In some cases, a bubble formed within microchannel **134** will remain small and will not affect any fluid flow or reaction occurring within microchannel **134**. When heat is applied to target zone **166**, however, the heat can cause the bubble to expand due to the large thermal expansion coefficient of the bubble.

(31) FIGS. 7A to 7E illustrate an example embodiment showing how controller **28** can detect unwanted objects such as bubbles and analyze a reaction even when bubbles are present. In some cases, unwanted objects may cause the reaction to fail completely and need to be rerun. A failure might occur if there are too many bubbles to compensate for. In FIGS. 7A to 7E, however, controller **28** can proceed with the reaction and analysis even with an air bubble present.

(32) FIG. 7A illustrates an example embodiment of target zone **166** before any fluid has entered the target zone. As illustrated, controller **28** has divided target zone **166** into a plurality of bins **200**. In the illustrated embodiment, controller **28** has divided target zone **166** into six rows (labeled A to F for ease of reference) and twelve columns (labeled 1 to 12 for ease of reference) to create seventy-two bins **200**, but those of ordinary skill in the art will recognize that any number of bins **200** can be created within a target zone **166**. It should be understood that the bins **200** are created virtually by controller **28**, and there are no physical barriers separating the bins **200** within target zone **166**.

(33) FIG. 7B illustrates target zone **166** after receiving a fluid sample but before a reaction has occurred. As illustrated, there are a minimal number of nucleic acid molecules **202** in the fluid sample before the reaction. In the illustrated embodiment, bins B7, C3, C10, D8 and E2 contain nucleic acid molecules **202**.

(34) FIG. 7C illustrates target zone **166** during a PCR. As illustrated, the molecules shown in bins B7, C3, C10, D8 and E2 in FIG. 7B have multiplied by diffusion. Clear colonies **204** of nucleic acid molecules **202** can be viewed because the diffusion speed of the molecules is slower than the speed required for the molecules to travel the distance between colonies. Bins A6, A7, B3, B6, B7, C3, C4, C9, C10, D2, D3, D7, D8, E1, E2, E3, E8 and E9 each contain nucleic acid molecules **202** in FIG. 7C.

(35) FIG. 7D illustrates bins **200** that have been selected by controller **28** to be used in calculations for amplification and melt curves and for the corresponding analysis. In the illustrated embodiment, the colonies **204** of nucleic acid molecules **202** are visualized by controller **28** based on localized increases in brightness in each bin. If a bin **200** reaches a threshold value for brightness, then that bin's fluorescence value is used in calculations for amplification and melt curves and for the corresponding analysis. If a bin **200** does not reach the threshold value for brightness, then that bin **200** is excluded from the calculations for amplification and melt curves and for the corresponding

analysis. In order to get a final value for fluorescence of the entire target zone **166**, the arithmetic mean is taken between all of the selected bins **200**. In the illustrated embodiment, bins A6, A7, B6, B7, C3, C9, C10, D2, D3, D7, D8 and E2 have been selected for use in calculations for amplification and melt curves and for the corresponding analysis, while bins B3, C4, E1, E3, E8 and E9 have been excluded even though they contain nucleic acid molecules because they do not meet the threshold for brightness.

(36) FIG. 7E illustrates how an unwanted object such as a bubble affects the analysis by controller **28**. In the illustrated embodiment, a bubble **206** has been detected by controller **28** in bins C3, C4, C5, C6, D3, D4, D5, D6, E3, E4, E5, E6, F3, F4 and F5. Even though bins C3 and D3 met the threshold value for brightness, controller **28** has determined that bins C3 and D3 should be excluded from the amplification and melt curves and corresponding analysis due to the presence of bubble **206**. Controller **28** has also determined that bins D2 and E2 should be given a lower weight during calculations due to their proximity to bubble **206**.

(37) Based on the above analysis, controller **28** can average the fluorescence values of the selected bins **200**. If controller **28** does not detect any bubbles in the images taken by camera imaging device **20**, then controller **28** calculates the fluorescence using the following equation:

$$(38) F = \frac{\text{Math.}_{i=1}^n x_i}{n},$$

wherein n is the number of bins **200** used in the calculation, and x.sub.i is the individual bins in the grid utilized by the algorithm.

(39) If one or more bubble is detected in target zone **166**, then controller **28** calculates the fluorescence using the following equation:

$$(40) F = \frac{\text{Math.}_{i=1}^n a_i x_i}{n},$$

where a.sub.i is the bin's weight, which is determined by the bin's proximity to the bubble. In an embodiment, a.sub.i should be greater than zero and less than or equal to one ($0 < a_{\text{sub}.i} \leq 1$).

(41) In an embodiment, the a.sub.t's may all have an equal weight of 1 when there are no bubbles present. In an embodiment, the a.sub.t's value can decrease proportionally as the bubble gets closer to the bin.

(42) In an embodiment, the amplification curve and melt curve are created by measuring the absolute value of the fluorescence of each pixel in successive images taken either during each cycle (for the amplification curve) or as temperature is slowly increased over time (for the melt curve). In the case of an amplification curve, which shows the incremental change in fluorescence during the amplification stage of a PCR, a PCR crossover threshold value can be determined, and controller **28** can look for when the fluorescence value has increased past a certain threshold value and when the first derivative of fluorescence with respect to cycle number is at a maximum. In the case of a melt curve, the incremental change in fluorescence can give controller **28** the melting temperature of the amplified DNA. Mathematically, controller **28** looks for when the first derivative of the fluorescence with respect to temperature is at a minimum, and the second derivative of the fluorescence with respect to temperature is equal to zero.

(43) The above detection and analysis is useful for a fluorescence based PCR or cytometry analysis, but those of ordinary skill in the art will recognize that controller **28** can be used for other purposes. In an embodiment, controller **28** can use camera imaging device **20** to perform a colorimetric analysis, which analyzes the concentration of a chemical element or chemical compound in a solution with the aid of a color reagent. Controller **28** can quantify the amount of protein present in the fluid sample by measuring the absorption spectra and comparing it with protein solutions of known concentration. In an embodiment, controller **28** can analyze and interpret the results of colorimetric protein assays, for example, a Bradford protein assay, a bichinchonin acid assay (BCA assay), and/or a Ponceau S dye assay.

(44) FIG. 8 illustrates an example embodiment of a control method that can be used by controller **28** to perform and analyze a reaction as described herein, and FIG. 9 illustrates an example embodiment of a controller **28** that can perform the method of FIG. 8. As illustrated, controller **28**

can include a processor **250** and a memory **252**, which can include a non-transitory computer readable medium. Memory **252** can include, for example, an input module **254**, a control module **256**, an analysis module **258**, and an output module **260**. Processor **250** can run the modules **252**, **254**, **256**, **258** in accordance with instructions stored on memory **252**. The broken lines in FIG. **8** illustrate the electrical connections between the modules **252**, **254**, **256**, **258** of controller **28** and various elements of device **10**. It should be understood by those of ordinary skill in the art that the illustrated modules and/or additional modules can be connected to the elements shown and/or additional elements.

(45) The process begins by loading a test card **100** and/or a fluid sample into device **10**. The fluid sample can be mixed with a reagent before injection into test card **100** and/or device **10**, or can be mixed with a reagent within mixing chamber **126** of test card **100**. In an embodiment, the reagent includes a PCR inhibitor-resistant polymerase along with a specific mixture of reverse transcriptase (in the case of RNA targets) and surfactants/dispersants to allow for rapid sample dispersion and lysing. In an embodiment, the reagent mix can include, for example, oligonucleotide primers, dNTP's DNA polymerase and other chemicals to assist the PCR. It is important to have a correct ratio of fluid sample to final PCR volume, because if the correct ratio is not maintained, the PCR will take too long or fail.

(46) Using user interface **60**, a user can then choose a positive/negative test to run on the fluid sample. In an embodiment, the user can cycle through a plurality of tests on display **62** using buttons **64** and choose one or more test to run. The plurality of tests can include, for example, a PCR analysis, a cytometry analysis and/or an enzyme-linked immunosorbent assay (ELISA) analysis. In an alternative embodiment, a plurality of different types of test cards **100** can be inserted into device **10**, with each test card **100** corresponding to one or more specific test to be run on a fluid sample, and controller **28** can determine which test(s) to run by detecting the type of test card **100** inserted into device **10** without further instruction by the user. In another alternative embodiment, fluid microchannel **134** can be incorporated into device **10** rather than test card **100**, and a user can choose a test to run on the fluid sample after the fluid sample is injected directly into device **10**.

(47) Input module **254** is configured to receive the user inputs inputted into user interface **60** and communicate the user inputs to control module **256**. Input module **254** can also receive additional information via user interface **60** and/or by the preprogramming of controller **28**, for example, (i) real-time PCR crossover threshold value information; (ii) maximum fluorescence information; and (iii) melting curve inflection temperature information. Other control parameters for the PCR reaction can also be given. This includes the PCR denaturing, elongation and annealing temperatures and dwell times.

(48) Once a test card **100** and/or fluid sample has been loaded into device **10**, control module **256** of controller **28** begins the control method at step **200** by causing fluid actuation source **26** to pull fluid through microchannel **134**. In the illustrated embodiment, fluid actuation source applies a negative pneumatic force to outlet port **130** via pneumatic tube **40** to pull fluid through microchannel **134**. In an alternative embodiment, fluid actuation source can include one or more other type of pump in fluid communication with microchannel **134**.

(49) After fluid actuation source **26** has been activated, but before the reaction begins, control module **256** at step **202** can verify that fluid is located within target zone **166** by monitoring the capacitance of microchannel **134** at one or more locations upstream and/or downstream of target zone **166**. If fluid is detected in microchannel **134** upstream and/or downstream of target zone **166**, control module **256** can verify that fluid is located within target zone **166**, activate light source **22** at step **204**, and begin the reaction at step **206**. Controller **28** can choose a portion of microchannel **134** to designate as target zone **166**, or target zone **166** can be predetermined.

(50) In the illustrated embodiment, control module **256** begins the reaction at step **206** by instructing power source **30** to send a current to electrodes located adjacent to target zone **166** via

electrical contact device **26** to cause the fluid within target zone **166** to be heated. As the fluid sample is heated, the nucleic acid molecules multiply by diffusion as explained above.

(51) At the same time that the fluid sample is being heated within target zone **166** so that the nucleic acid molecules multiply by diffusion, control module **256** at step **208** can cause camera imaging device **20** to record a plurality of images of the reaction within target zone **166**. The plurality of images can then be sent to analysis module **208** for analysis at step **210**. In an embodiment, test card **100** includes a transparent material that allows images to be taken of target zone **166** of fluid microchannel **134** even though a layer of polymer material is located between camera imaging device **20** and fluid microchannel **134**.

(52) At step **210**, analysis module **258** analyzes the images taken by camera imaging device **20** to determine whether the fluid sample tests positive or negative for a bacteria or virus. The type of analysis performed by analysis module at step **210** will depend on the type of test being run on the fluid sample.

(53) If the assay being run on the fluid sample is a PCR analysis, then analysis module **258** can analyze the images as discussed above by measuring fluorescence based on detected wanted and unwanted objects.

(54) If the assay being run on the fluid sample is a cytometry analysis, then analysis module **258** can also analyze the images as discussed above by measuring fluorescence based on detected wanted and unwanted objects. The cytometry analysis can differ from the PCR analysis, for example, because the fluid in target zone **166** does not need to be heated to multiply molecules by diffusion, so step **206** can be skipped. With a cytometry analysis, analysis module **258** can analyze the fluid sample within target zone **166**, for example, by analyzing cell size, cell count, cell morphology (shape and structure), cell cycle phase, DNA content, and the existence or absence of specific proteins on cell surfaces. In an embodiment, controller **28** can use various different fluorophores for flow cytometry. In an embodiment, to detect specific proteins on cell surfaces, specifically designed fluorophores which bind to those proteins are mixed into the sample to cause the proteins of interest to fluoresce.

(55) If the assay being run on the fluid sample is an ELISA analysis, then again the fluid in target zone **166** does not need to be heated to multiply molecules by diffusion, so step **206** can be skipped. With an ELISA analysis, analysis module **258** can analyze the fluid sample within target zone **166**, for example, by measuring the concentration of an analyte in the fluid sample using a colorimetric analysis. In an embodiment, controller **28** can measure the amount of incident light scattered on the ELISA target zone to determine the concentration of an analyte. The method of measurement is the same as the turbidity measurement.

(56) In an embodiment, controller **28** can perform genotyping tests as an extension a PCR.

(57) At step **212**, analysis module **258** determines based on the analysis whether the fluid sample has tested positive or negative for a bacteria or virus. The results of the analysis are then displayed on user interface **60** by output module **260**. In an embodiment, a simple “POSITIVE” or “NEGATIVE” indication can be displayed on user interface **60** to inform whether the fluid sample has tested positive or negative for a bacteria or virus. In another embodiment, user interface **60** can display the results for more than one bacteria or virus, or can display specifics such as cell size, cell count, cell morphology (shape and structure), cell cycle phase, DNA content, and the existence or absence of specific proteins on cell surfaces. In an embodiment, controller **28** can display viral titer in the case of a PCR reaction, or can display protein concentration and DNA concentration.

(58) In an embodiment, the result of the analysis can be saved in a memory module of memory **252**, so that the results can be reviewed at a later time. If the result is saved, the result should be encoded to protect the anonymity of the patient. In another embodiment, one or more encoded results can be wirelessly transmitted for review at a location remote from diagnostic system **1**.

(59) In an embodiment, device **10** can include a global positioning system (GPS) sensor, and can record the result of the test along with a GPS sensor reading at the time of the test. Controller **28**

can then aggregate a plurality of tests to determine viruses or bacteria that are more prevalent in one area as opposed to another. In this embodiment, controller **28** does not need to save any patient information, and only needs the GPS location and the number of positive and negative test results at the location to determine the prevalence of the virus or bacteria at the location. The results can be used by a health organization to treat an area with appropriate medication for a prevalent virus or bacteria.

(60) In another embodiment, the user can program the location into device **10** before, during or after running a plurality of tests, and controller **28** can aggregate the plurality of tests to determine viruses or bacteria that are more or less prevalent in the programmed area.

(61) In an embodiment, controller **28** can determine whether one or more of yersina pestis, *brucella*, alphavirus, dengue virus and/or variola virus is present in the fluid sample.

(62) The diagnostic system **1** disclosed herein can detect, for example, lentiviruses (ssRNA) and adenoviruses (dsDNA) in whole blood, as well as other infectious agents. Exemplary infectious agents which can be detected by the system **1** disclosed herein include, but are not limited to, bacterial pathogens, viral pathogens, fungal pathogens, and/or parasitic pathogens.

(63) Exemplary non-limiting bacterial pathogens include *Bacillus* (e.g., *Bacillus anthracis*, *Bacillus cereus*), *Bartonella* (e.g., *Bartonella henselae*, *Bartonella quintana*), *Bordatella* (e.g., *Bordatella pertussis*), *Borrelia* (e.g., *Borrelia burgdorferi*, *Borrelia garinii*, *Borrelia afzelii*, *Borrelia recurrentis*), *Brucella* (e.g., *Brucella abortus*, *Brucella canis*, *Brucella melitensis*, *Brucella suis*), *Campylobacter* (e.g., *Campylobacter jejuni*), *Chlamydia* (e.g., *Chlamydia pneumoniae*, *Chlamydia trachomatis*), *Chlamydophila* (e.g., *Chlamydophila psittaci*), *Clostridium* (e.g., *Clostridium botulinum*, *Clostridium difficile*, *Clostridium perfringens*, *Clostridium tetani*), *Corynebacterium* (e.g., *Corynebacterium diphtheriae*), *Enterococcus* (e.g., *Enterococcus faecalis*, *Enterococcus faecium*), *Escherichia* (e.g., *Escherichia coli*), *Francisella* (e.g., *Francisella tularensis*), *Haemophilus* (e.g., *Haemophilus influenzae*), *Helicobacter* (e.g., *Helicobacter pylori*), *Legionella* (e.g., *Legionella pneumophila*), *Leptospira* (e.g., *Leptospira interrogans*, *Leptospira santarosai*, *Leptospira weilii*, *Leptospira noguchii*), *Listeria* (e.g., *Listeria monocytogenes*), *Mycobacterium* (e.g., *Mycobacterium leprae*, *Mycobacterium tuberculosis*, *Mycobacterium ulcerans*), *Mycoplasma* (e.g., *Mycoplasma pneumoniae*), *Neisseria* (e.g., *Neisseria gonorrhoeae*, *Neisseria meningitidis*), *Pseudomonas* (e.g., *Pseudomonas aeruginosa*), *Rickettsia* (e.g., *Rickettsia rickettsii*), *Salmonella* (e.g., *Salmonella typhi*, *Salmonella typhimurium*), *Shigella* (e.g., *Shigella sonnei*), *Staphylococcus* (e.g., *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Staphylococcus saprophyticus*), *Streptococcus* (e.g., *Streptococcus agalactiae*, *Streptococcus pneumoniae*, *Streptococcus pyogenes*), *Treponema* (e.g., *Treponema pallidum*), *Ureaplasma* (e.g., *Ureaplasma urealyticum*), *Vibrio* (e.g., *Vibrio cholerae*), and *Yersinia* (e.g., *Yersinia pestis*, *Yersinia enterocolitica*, *Yersinia pseudotuberculosis*).

(64) Exemplary non-limiting viral pathogens include Adenoviridae (e.g., adenovirus), Herpesviridae (e.g., herpes simplex virus type 1 and type 2, varicella-zoster virus, cytomegalovirus, Epstein-Barr virus, human herpesvirus type 8), Papillomaviridae (e.g., human papillomavirus), Polyomaviridae (e.g., BK virus, JC virus), Poxviridae (e.g., smallpox), Hepadnaviridae (e.g., hepatitis B virus), Parvoviridae (e.g., human bocavirus, parvovirus B19), Astroviridae (e.g., human astrovirus), Caliciviridae (e.g., Norwalk virus), Picornaviridae (e.g., Coxsackievirus, hepatitis A virus, poliovirus, rhinovirus); Coronaviridae (e.g., severe acute respiratory syndrome virus, Middle East respiratory syndrome virus), Flaviviridae (e.g., hepatitis C virus, yellow fever virus, dengue virus, West Nile virus), Togaviridae (e.g., rubella virus), Hepeviridae (e.g., hepatitis E virus), Retroviridae (e.g., lentiviruses, human immunodeficiency virus); Orthomyxoviridae (e.g., influenza virus), Arenaviridae (e.g., Guanarito virus, Junin virus, Lassa virus, Machupo virus, Sabiá virus), Bunyaviridae (e.g., Crimean-Congo hemorrhagic fever virus), Filoviridae (e.g., Ebola virus, Marburg virus), Paramyxoviridae (e.g., measles virus, mumps virus, parainfluenza virus, respiratory syncytial virus, human metapneumonia virus, Hendra virus, Nipah virus),

Phabdoviridae (e.g., rabies virus), Reoviridae (e.g., rotavirus, orbivirus, coltivirus, Banna virus), and unassigned viruses (e.g., Hepatitis D virus).

(65) Exemplary non-limiting fungal pathogens include *Candida* (e.g., *Candida albicans*), *Aspergillus* (e.g., *Aspergillus fumigatus*, *Aspergillus flavus*), *Cryptococcus* (e.g., *Cryptococcus neoformans*, *Cryptococcus laurentii*, *Cryptococcus gattii*), *Histoplasma* (e.g., *Histoplasma capsulatum*), *Pneumocystis* (e.g., *Pneumocystis jirovecii*, *Pneumocystis carinii*), *Stachybotrys* (e.g., *Stachybotrys chartarum*).

(66) Exemplary non-limiting parasitic pathogens include *acanthamoeba*, *anisakis*, *Ascaris lumbricoides*, botfly, *Balantidium coli*, bedbugs, *Cestoda* (tapeworm), chiggers, *Cochliomyia hominivorax*, *Entamoeba histolytica*, *Fasciola hepatica*, *Giardia lamblia*, hookworms, *Leishmania*, *Linguatula serrata*, liver flukes, *Loa loa*, *Paragonimus*—lung fluke, pinworm, *Plasmodium falciparum*, *Schistosoma*, *Strongyloides stercoralis*, mites, tapeworms, *Toxoplasma gondii*, *Trypanosoma*, whipworms, and *Wuchereria bancrofti*.

(67) It should be understood that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and modifications can be made without departing from the spirit and scope of the present subject matter and without diminishing its intended advantages. It is therefore intended that such changes and modifications be covered by the appended claims.

(68) Unless otherwise indicated, all numbers expressing quantities of ingredients, properties such as molecular weight, reaction conditions, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the present disclosure. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

(69) The terms “a” and “an” and “the” and similar referents used in the context of the disclosure (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g. “such as”) provided herein is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the disclosure.

(70) The use of the term “or” in the claims is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.”

(71) Groupings of alternative elements or embodiments of the disclosure disclosed herein are not to be construed as limitations. Each group member may be referred to and claimed individually or in any combination with other members of the group or other elements found herein. It is anticipated that one or more members of a group may be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all

Markush groups used in the appended claims.

(72) Preferred embodiments of the disclosure are described herein, including the best mode known to the inventors for carrying out the disclosure. Of course, variations on those preferred embodiments will become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventor expects those of ordinary skill in the art to employ such variations as appropriate, and the inventors intend for the disclosure to be practiced otherwise than specifically described herein. Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

(73) Specific embodiments disclosed herein may be further limited in the claims using consisting of or consisting essentially of language. When used in the claims, whether as filed or added per amendment, the transition term “consisting of” excludes any element, step, or ingredient not specified in the claims. The transition term “consisting essentially of” limits the scope of a claim to the specified materials or steps and those that do not materially affect the basic and novel characteristic(s). Embodiments of the disclosure so claimed are inherently or expressly described and enabled herein.

(74) Further, it is to be understood that the embodiments of the disclosure disclosed herein are illustrative of the principles of the present disclosure. Other modifications that may be employed are within the scope of the disclosure. Thus, by way of example, but not of limitation, alternative configurations of the present disclosure may be utilized in accordance with the teachings herein. Accordingly, the present disclosure is not limited to that precisely as shown and described.

Claims

1. A device for analyzing a fluid sample, the device comprising: a housing defining a slot sized to receive a test card containing the fluid sample; a camera imaging device positioned within the housing and configured to record an image of a target zone on the test card; a light source positioned within the housing and configured to illuminate at least the target zone on the test card; and a controller positioned within the housing communicatively coupled to the camera imaging device and the light source, the controller comprising: a memory device comprising a non-transitory computer readable medium and storing instructions; and a processor communicatively coupled to the memory device, wherein the processor is operable to execute the instructions stored on the memory device to: activate the light source after detecting the test card was inserted through the slot, control the camera imaging device to record at least one image of the target zone on the test card, analyze the at least one image to distinguish wanted objects from unwanted objects within the at least one image, and determine, from the analysis, whether the fluid sample tests positive or negative for a bacterium or a virus based on the wanted objects, wherein the wanted objects include at least one of nucleic acid molecules or cells, and wherein the unwanted objects include objects that do not have at least one of an average size or an average shape of the at least one of nucleic acid molecules or cells.
2. The device of claim 1, wherein the processor is operable to execute the instructions stored on the memory device to determine whether the fluid sample has tested positive or negative for the bacterium or the virus by calculating a mean fluorescence value for at least one area that includes the wanted objects.
3. The device of claim 1, wherein the processor is operable to execute the instructions stored on the memory device to determine whether the fluid sample has tested positive or negative for the bacterium or the virus by calculating a mean fluorescence value for at least one area that excludes the unwanted objects.
4. The device of claim 1, wherein the processor is operable to execute the instructions stored on the

memory device to determine whether the fluid sample has tested positive or negative for the bacterium or the virus by analyzing at least one of a size of the wanted objects, a count of the wanted objects, a morphology of the wanted objects, a cycle phase of the wanted objects, DNA content of the wanted objects, or an existence or absence of specific proteins on surfaces of the wanted objects.

5. The device of claim 1, wherein the processor is operable to execute the instructions stored on the memory device to determine whether the fluid sample has tested positive or negative for the bacterium or the virus by: counting colonies that are shown within the at least one image; comparing a number of counted colonies to a specified titer value; and determining the fluid sample has tested positive for the bacterium or the virus when the number of counted colonies exceed the specified titer value.

6. The device of claim 1, wherein the processor is further operable to execute the instructions stored on the memory device to control the device to perform a cytometry analysis.

7. The device of claim 6, wherein the memory device stores a plurality of preprogrammed analyses that can be performed, the plurality of preprogrammed analyses including the cytometry analysis and a polymerase chain reaction (“PCR”) analysis, and wherein the processor is operable to execute the instructions stored on the memory device to operate according to the preprogrammed cytometry analysis.

8. The device of claim 1, wherein the processor is operable to execute the instructions stored on the memory device to report an inconclusive test when the processor identifies at least one unwanted object in the at least one image.

9. The device of claim 1, further comprising a display device coupled to the housing and communicatively coupled to the processor, wherein the processor is further operable to execute the instructions stored on the memory device to control the display device to display an indication whether the fluid sample tests positive or negative for the bacterium or the virus.

10. The device of claim 1, wherein the processor is operable to execute the instructions stored on the memory device to control the camera imaging device to record a plurality of images, wherein the unwanted objects include air bubbles, and wherein the processor is operable to execute the instructions stored on the memory device to distinguish the air bubbles by identifying objects that expand in size during the recording of the plurality of images.

11. The device of claim 1, wherein the processor is further operable to execute the instructions stored on the memory device to distinguish the wanted objects from the unwanted objects by comparing a size or shape of objects in the at least one image to an average size or shape of blood cells.

12. The device of claim 1, further comprising a heater positioned within the housing and electrically coupled to the processor, wherein the processor is further operable to execute the instructions stored on the memory device to activate the heater after the light source is activated.

13. The device of claim 1, wherein the target zone on the test card includes a single zone.

14. The device of claim 1, wherein the target zone on the test card includes a plurality of zones.

15. The device of claim 1, wherein the light source is configured to emit fluorescence excitation light.

16. The device of claim 1, wherein the processor is operable to execute the instructions stored on the memory device to perform at least one of a polymerase chain reaction (“PCR”) analysis, a cytometry analysis, or an enzyme-linked immunosorbent assay (“ELISA”) analysis to determine whether the fluid sample tests positive or negative for the bacterium or the virus based on the wanted objects.

17. The device of claim 1, further comprising a fluid actuation source positioned within the housing and configured to move the fluid sample to at least the target zone on the test card, wherein the processor is further operable to execute the instructions stored on the memory device to activate the fluid actuation source after detecting the test card was inserted through the slot.

18. The device of claim 17, wherein the processor is further operable to execute the instructions stored on the memory device to: measure a capacitance of a microchannel on the test card before or after the target zone; and at least one of activate the light source or control the camera imaging device to record the at least one image of the target zone after determining the measured capacitance is indicative that at least some of the fluid sample is in the target zone.
19. The device of claim 1, wherein the processor is further operable to execute the instructions stored on the memory device to locate the target zone with the at least one image, and wherein the processor is operable to determine whether the fluid sample tests positive or negative for the bacterium or virus to facilitate treatment of a bacterial or viral infection in a host.
20. The device of claim 1, wherein the processor is further operable to execute the instructions stored on the memory device to determine, from the analysis, at least one of a cell size, a cell count, a cell morphology, a cell cycle phase, a DNA content, the existence or absence of specific proteins on cell surfaces, viral titer, protein concentrations, and/or DNA concentrations.
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